B3-EMG-Workshop

2026.03.23

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D2

Graduate School of Information Science and Engineering

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The World of EMG

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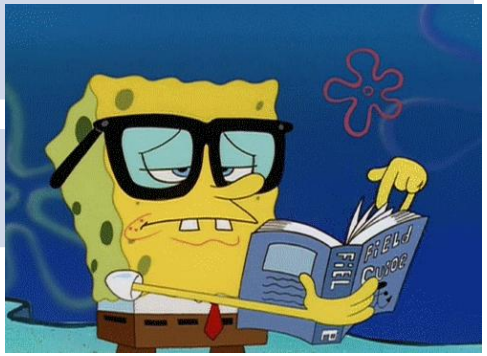

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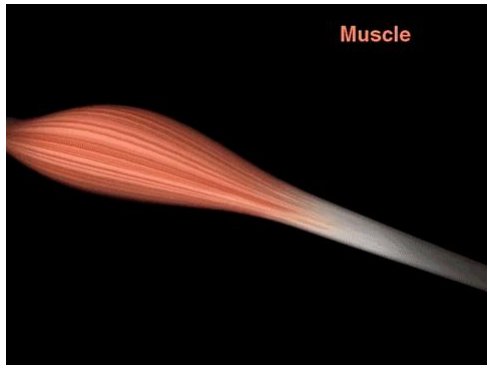
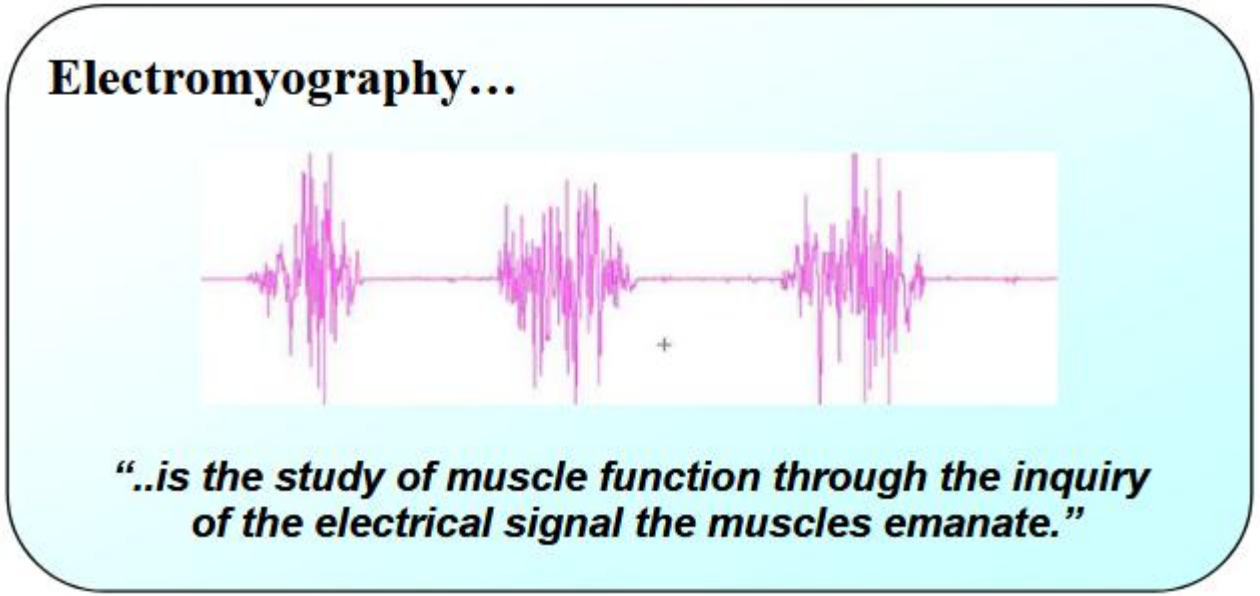
What is EMG (Electromyography)?

- ***Objective:*** *To provide a basic understanding of Electromyography (EMG).*
  - **Definition:** EMG is a technique used to record electrical activity generated by muscle contractions.





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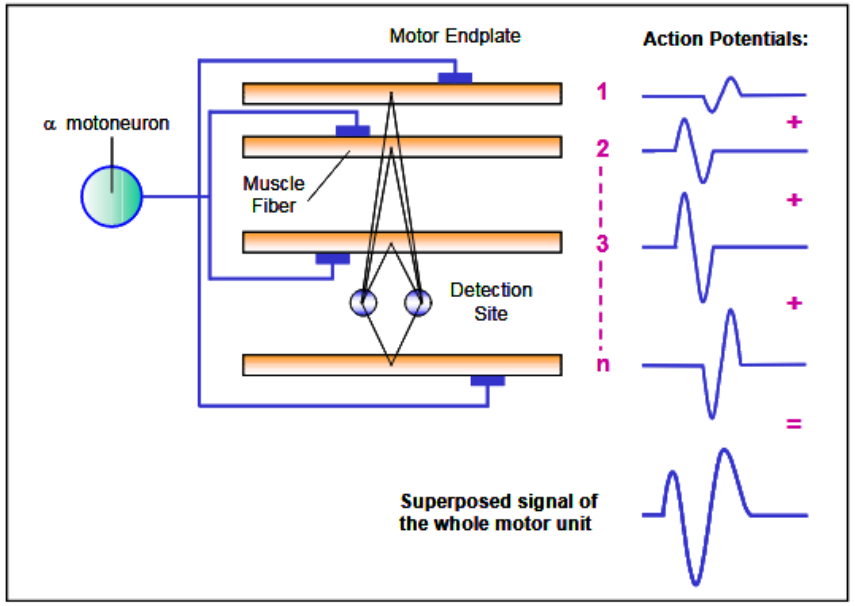

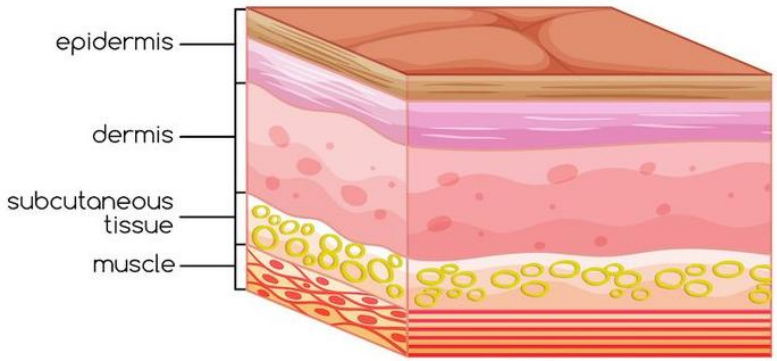

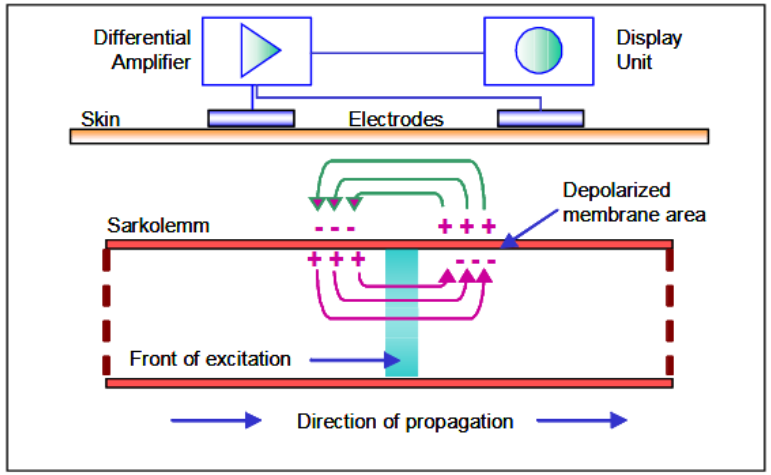

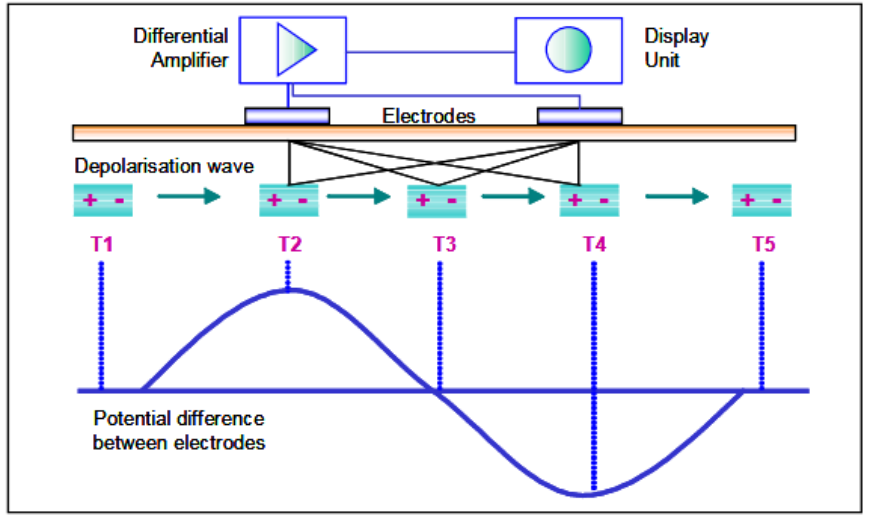

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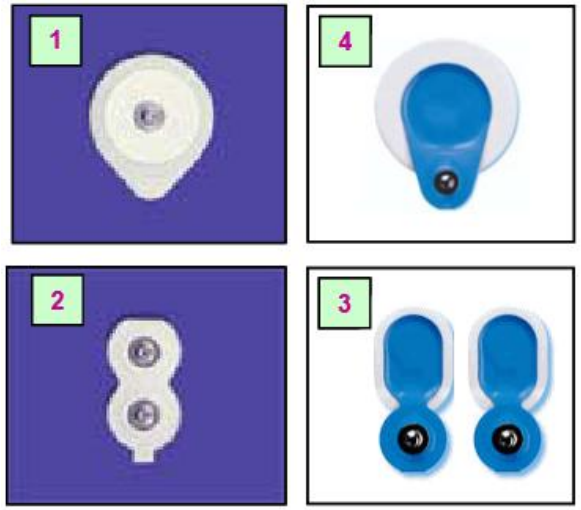
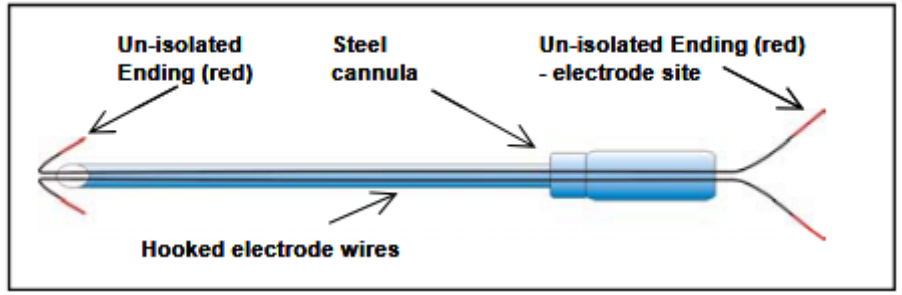
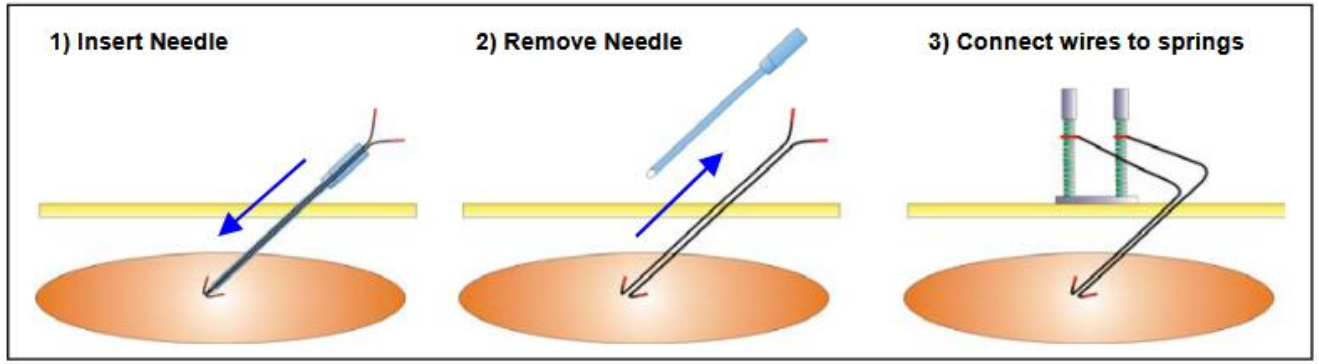
- **How EMG works:** It uses electrodes placed on the skin or inside the muscles to detect electrical signals produced by nerve and muscle activity.
- **Types of EMG:**
  - **Surface EMG (sEMG):** Non-invasive, electrodes are placed on the skin surface.
  - **Intramuscular EMG:** Invasive, electrodes are inserted into the muscles.







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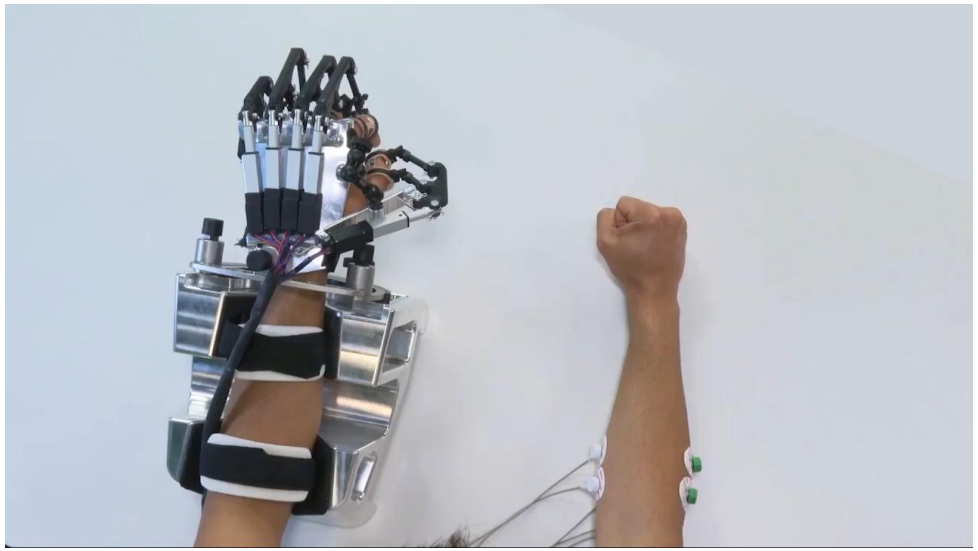
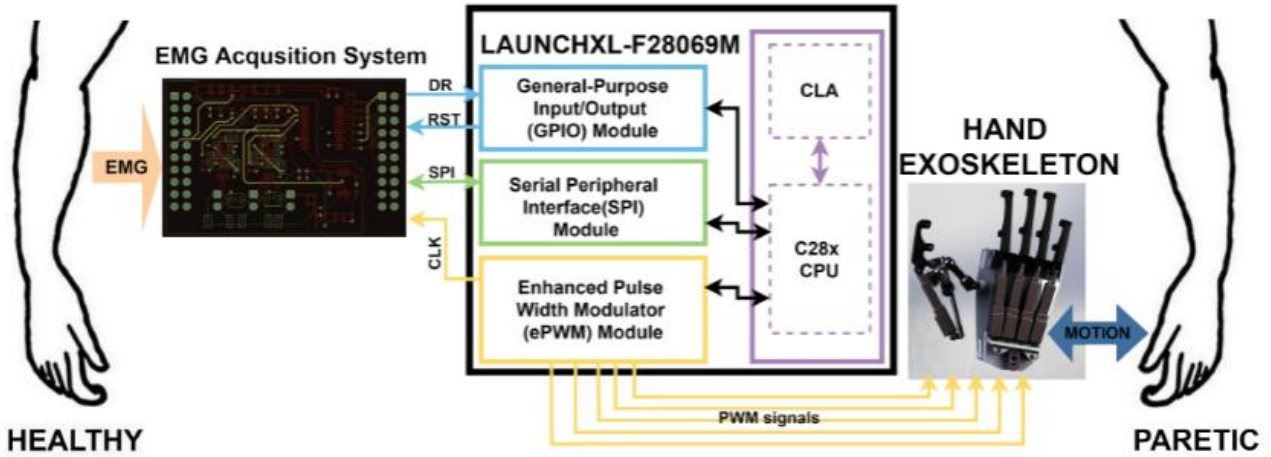
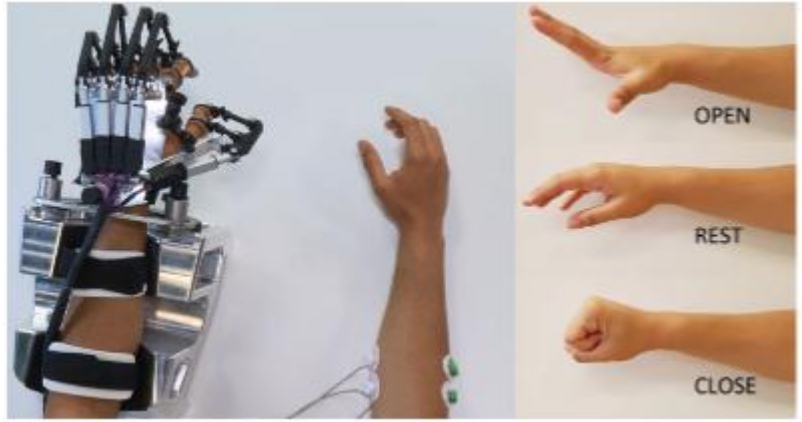
What is EMG Used For?

- ***Objective:*** *Providing an overview of EMG Applications in Real Life.*
- **Research and Use Cases:**
  - **Medical and Rehabilitation:**
    - Diagnosis of neuromuscular diseases such as ALS and muscular dystrophy.
    - Post-stroke rehabilitation to restore muscle function.







RobHand: Hand Exoskeleton With Real-Time EMG-Driven Embedded Control. Quantifying Hand Gesture Recognition Delays for Bilateral Rehabilitation

A. Cisnal et al. | Date of Publication: 06 October 2021 | IEEE Access | DOI: 10.1109/ACCESS.2021.3118281

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  - **Prosthetic and Robotic Control:**
    - Controlling robotic hands/legs based on user movement intentions.

Myo Armbands Used to Control Prosthetic Arm (Johnny Matheny picks up a ball using the Myo-controlled arm

Michelle Starr | Date of Publication: 18 January 2016 | CNET News

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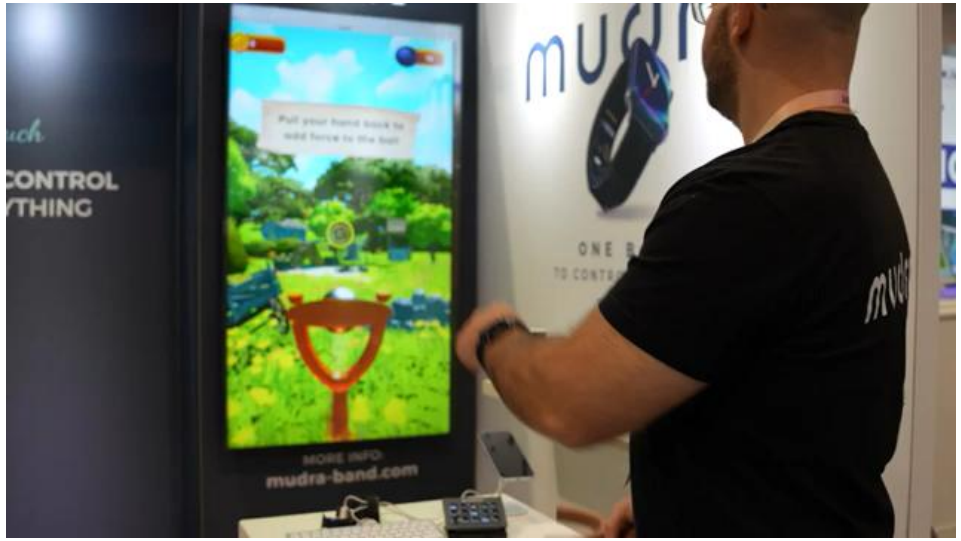
  - **Sports and Athletic Performance:**
    - Motion analysis to improve training efficiency or injury recovery.
  - **Human-Computer Interaction (HCI):**
    - Control of wearable devices, games, or augmented reality (AR) technology.



Introducing a New Revolutionary "Spatial Depth" Control in Latest Air Mouse Innovation Wearable Devices – Youtube | Date of Publication: 07 February 2024 | Mudra-Band

Meta Reality Labs: Wristband for AR r/AugmentedReality – Youtube | Date of Publication: 19 March 2021 | Meta Facebook

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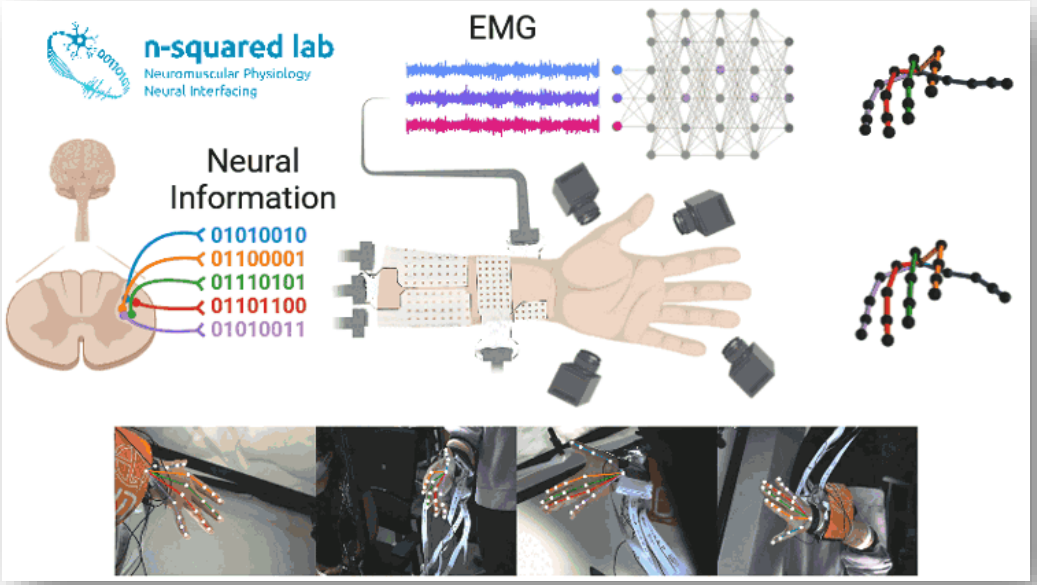
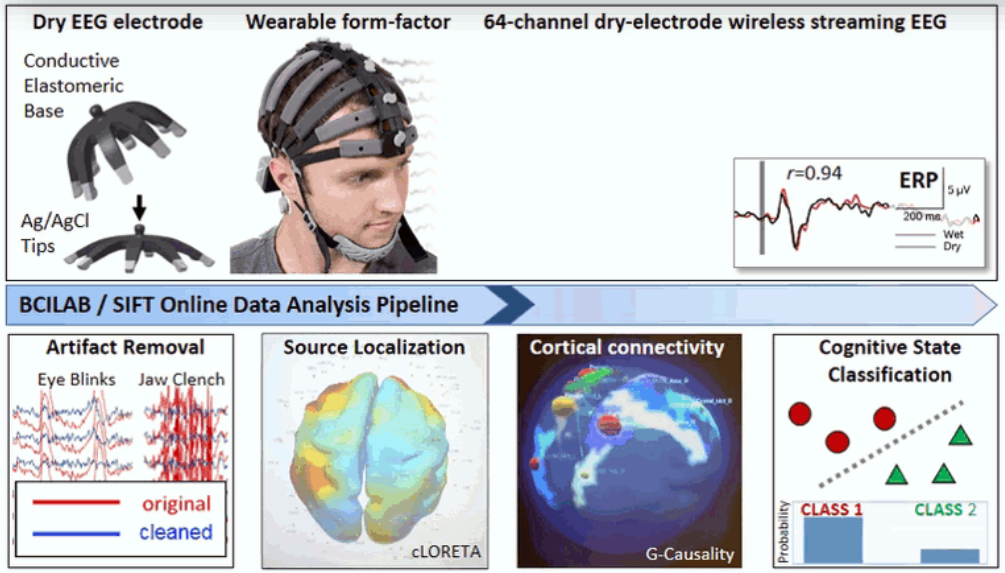
Why Use EMG Instead of EEG?

- ***Objective:*** *Comparing EMG with other technologies to identify EMG's specific advantages.*
  - **EMG vs EEG (Electroencephalography):**
    - **EMG:** Focuses on local muscle activity (specific movements).
    - **EEG:** Focuses on global brain activity (general instructions or intentions).
  - **Advantages of EMG:**
    - Higher spatial resolution for more specific movements.
    - More responsive and faster for device control.
  - **When to Use EEG:**
    - To detect movement intentions without direct muscle contraction, for example, in patients with complete paralysis.



Learning a Hand Model from Dynamic Movements Using High-Density EMG and Convolutional Neural Networks

Raul C. Simpetru, et al. | Date of Publication: 29 November 2024 | IEEE Transactions on Biomedical Engineering | DOI: 10.1109/TBME.2024.3432800



Real-time Neuroimaging and Cognitive Monitoring Using Wearable Dry EEG

IEEE EMBS | Date of Publication: 22 October 2015 | IEEE Transactions on Biomedical Engineering | [Website]

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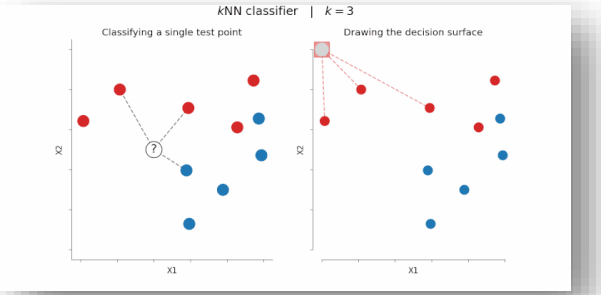
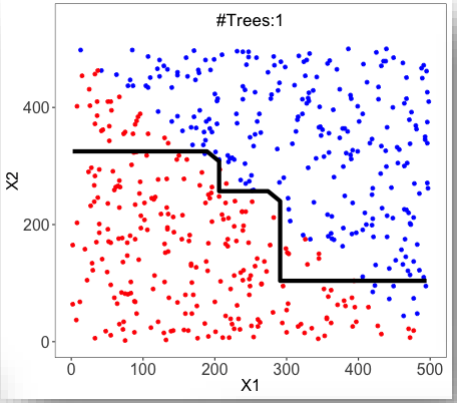
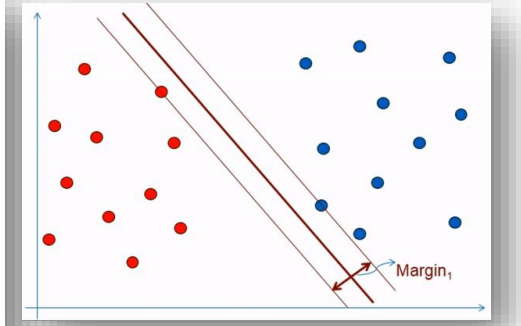
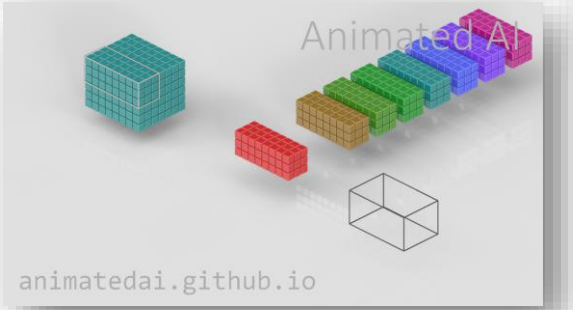
Advancements in EMG Motion Recognition Techniques

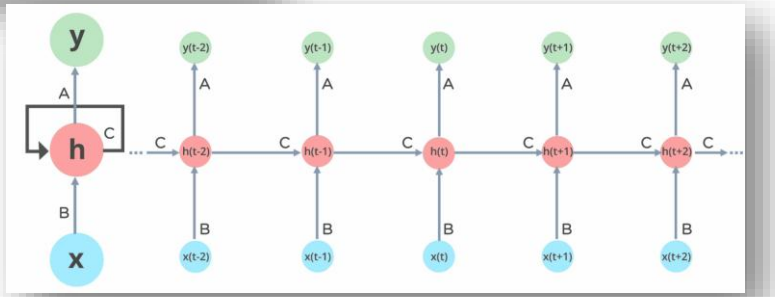
- ***Objective:*** *Discuss the latest techniques.*
  - **Classical Techniques:**
    - K-Nearest Neighbor (KNN), Decision Tree, and SVM (Support Vector Machine).
  - **Modern Techniques:**
    - CNN (Convolutional Neural Networks) and RNN (Recurrent Neural Networks) for sequential signal processing.
    - Transformers and generative models like ChatEMG for synthetic data and signal personalization.
  - **Practical Applications:**
    - Robotic hand control for individuals without limbs.
    - Prosthetic control in other body areas with active muscles (e.g., shoulders, face, etc.).

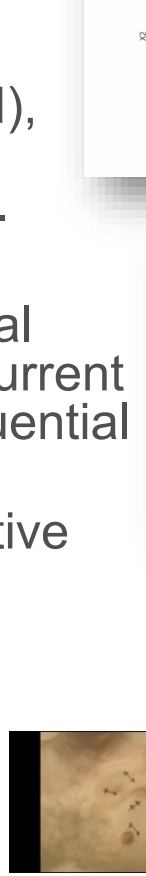


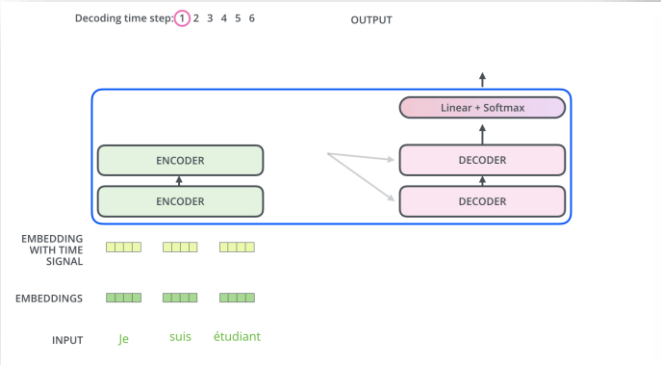


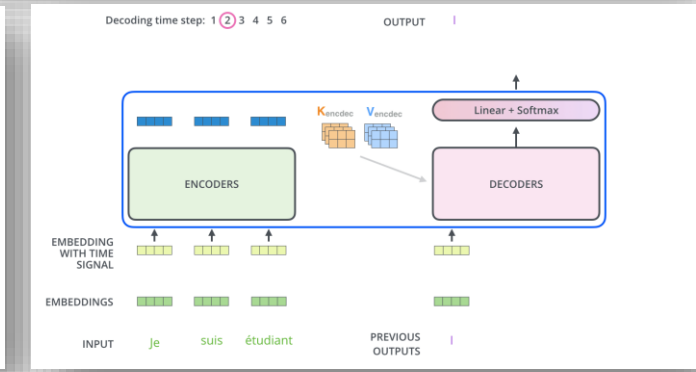












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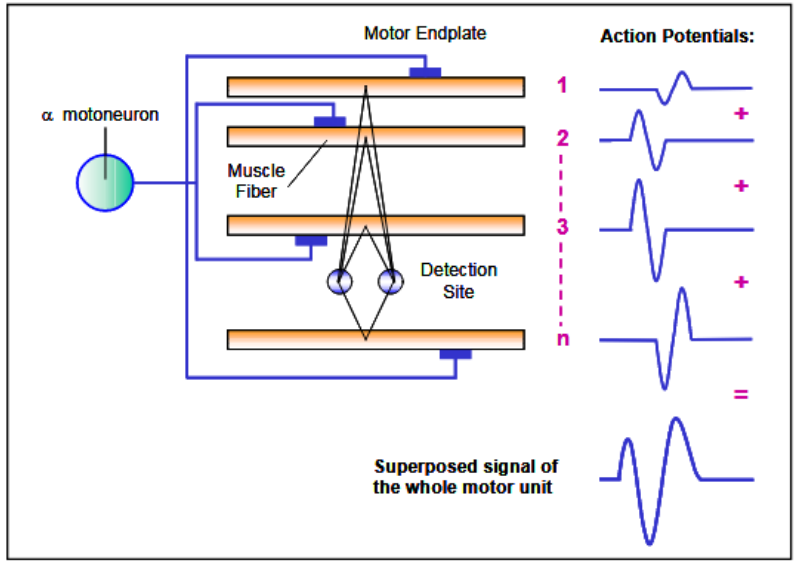
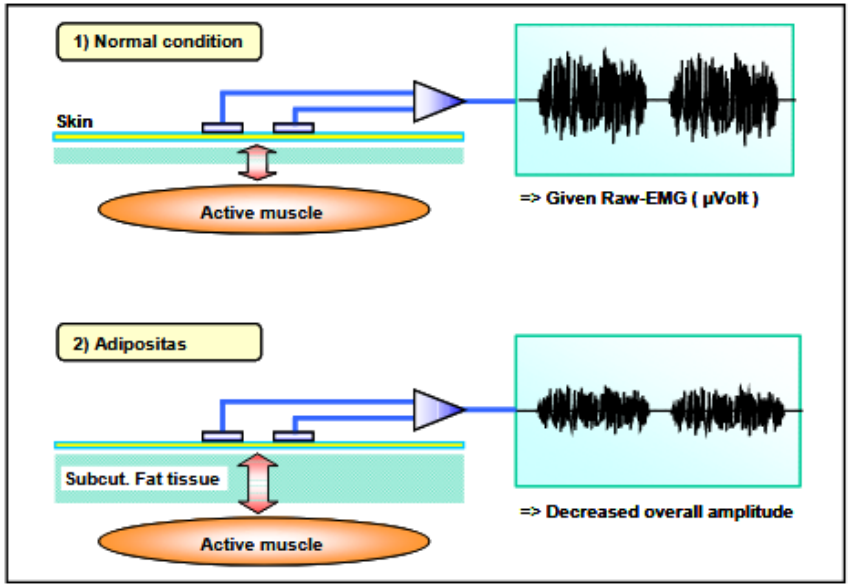
Specific Challenges of EMG

- ***Objective:*** *Identifying relevant technical challenges to highlight the importance of new research.*
- **Specific Challenges:**
  - **Signal Variability:**
    - **Inter-Individual:** Physiological differences such as muscle size, age, and skin thickness affect signal quality.
    - **Intra-Individual:** The same movement may produce different patterns due to muscle fatigue or changes in electrode positioning.





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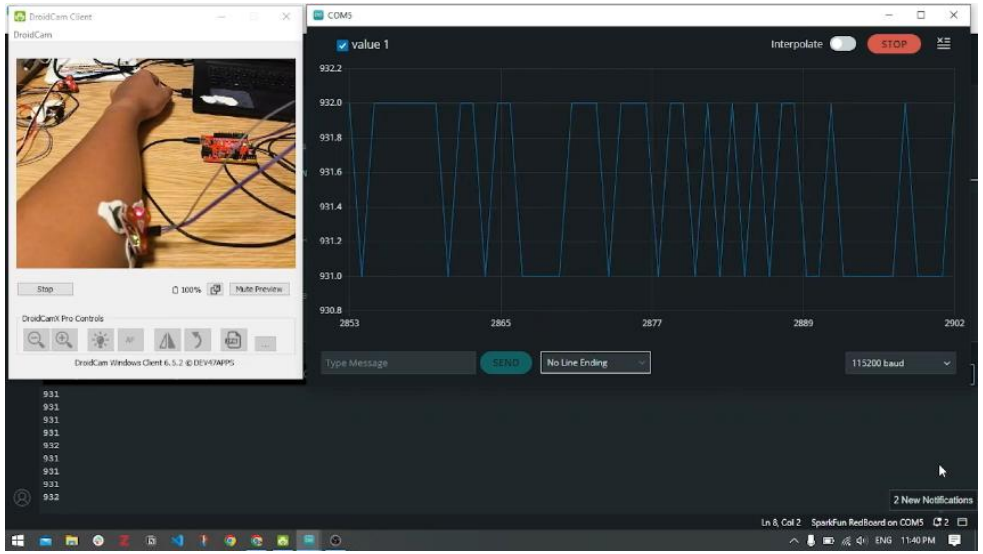
    - **Intra-Individual:** The same movement may produce different patterns due to muscle fatigue or changes in electrode positioning.
  - **Noise and Interference:**
    - Signals are easily disturbed by electrical noise (50-60 Hz) and electrode movement.
  - **Non-Stationary Signals:**
    - EMG signals change over time and with device usage.
  - **Model Generalization Limitations:**
    - Models trained under specific conditions or for certain individuals often fail to generalize to other conditions or patients.



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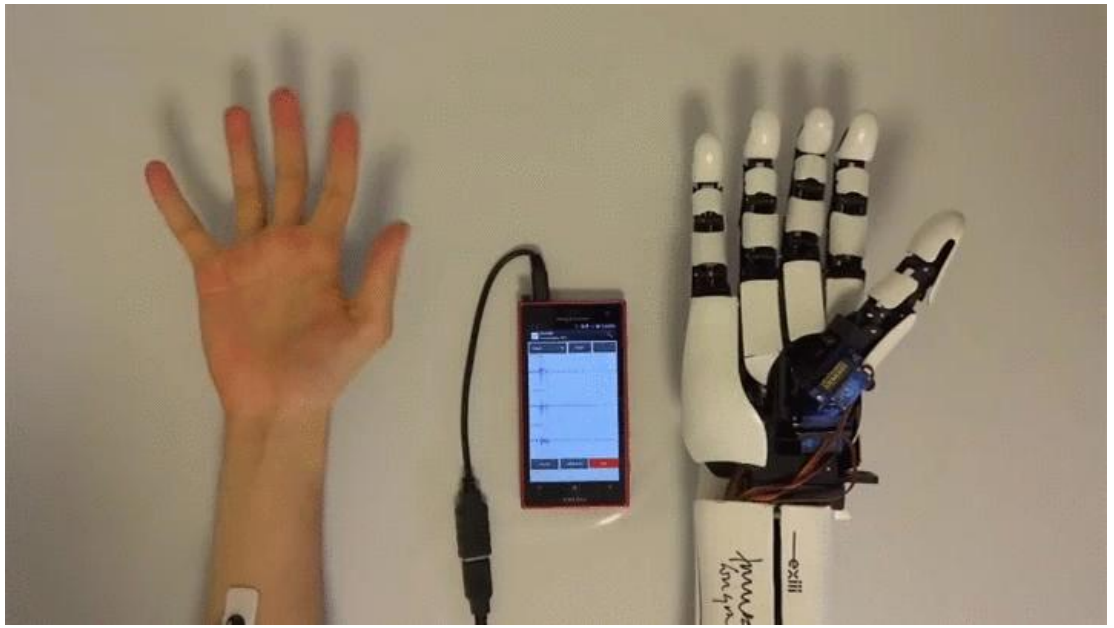
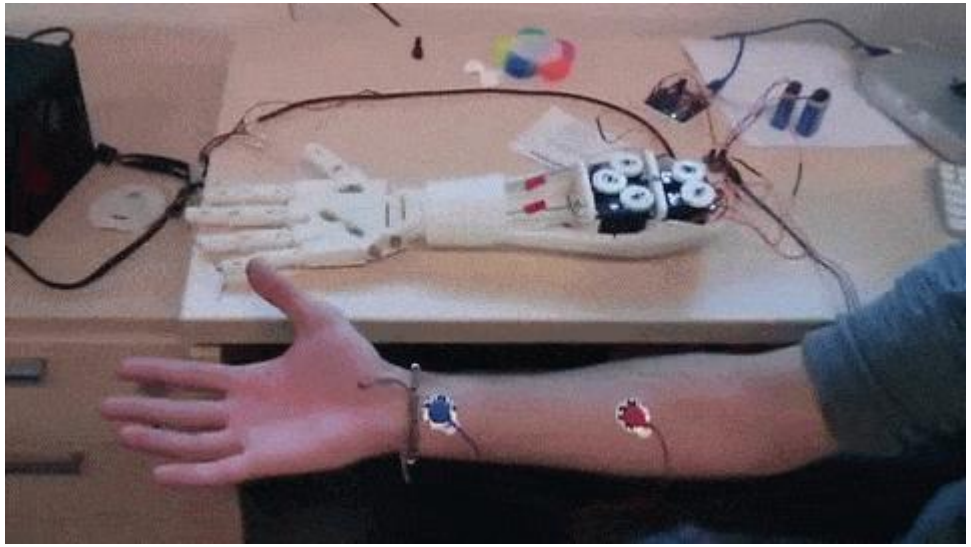
Advantages and Disadvantages of EMG

- ***Objective:*** *Managing participant expectations before introducing the paper's solution.*
- **Advantages:**
  - High resolution for detecting small movements.
  - Easy integration with wearable and robotic devices.
  - Non-invasive (for sEMG).
- **Disadvantages:**
  - Sensitive to noise.
  - Requires recalibration for new users or sessions.
  - Affected by physiological and environmental conditions.





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The 6 Points from The World of EMG

1. Physiological Variability Affects EMG Signal Quality
2. Intra-Subject Variability Due to Muscle Fatigue and Electrode Position
3. Signal Changes Over Time and Device Usage
4. Non-Generalizable Models Struggle to Adapt to New Subjects or Conditions
5. Sensitivity to Noise Reduces Detection Accuracy
6. Frequent Recalibration Needed for New Users or Sessions

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ありがとうございます。

THANK YOU

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